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(54) **REFRIGERATION SYSTEM AND METHOD
ENABLING DEMAND FLEXIBILITY**

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ABSTRACT

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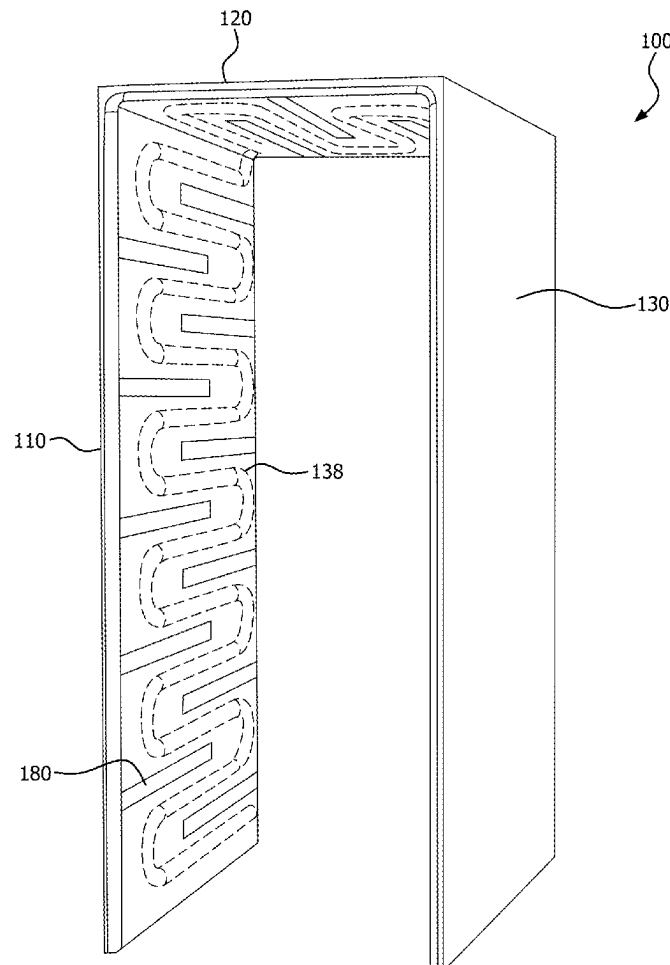
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A refrigeration system includes one or multiple refrigeration compartments, one or multiple phase change evaporators, one or multiple fans, a compressor and a condenser. The phase change evaporators and fans are within the refrigeration compartments, and comprises an open cell heat conductive foam having a phase change material embedded within the cells of the open cell heat conductive foam. A refrigerant conduit is provided and is in thermal contact with the open cell heat conductive foam, and contains a refrigerant fluid and communicates with a refrigeration cycle conduit which circulates the refrigerant in a refrigeration cycle. The fans are provided to increase heat transfer between the phase change evaporators and the refrigeration compartments. A method of refrigerating a refrigeration compartment and a phase change evaporator are disclosed. A method for refrigerating a refrigeration compartment, and an evaporator for a refrigeration system are also disclosed.



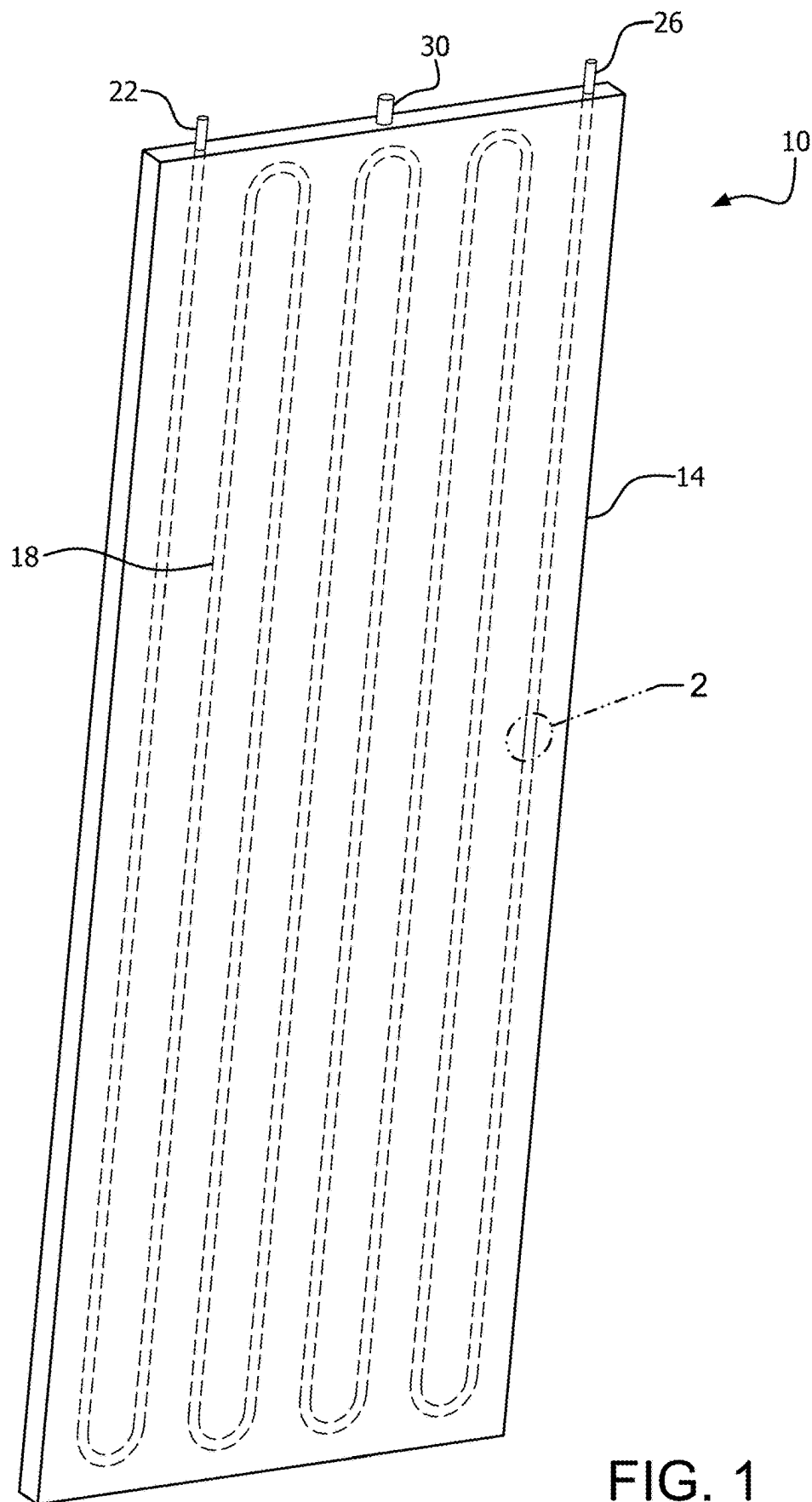
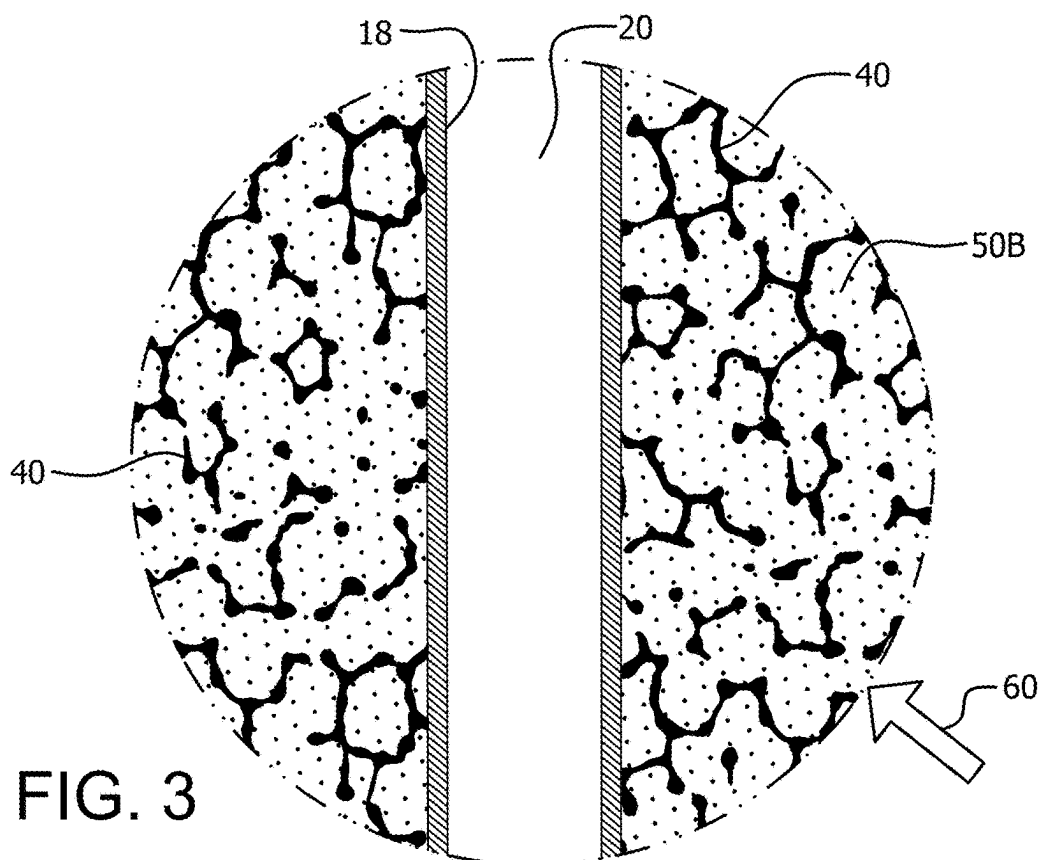
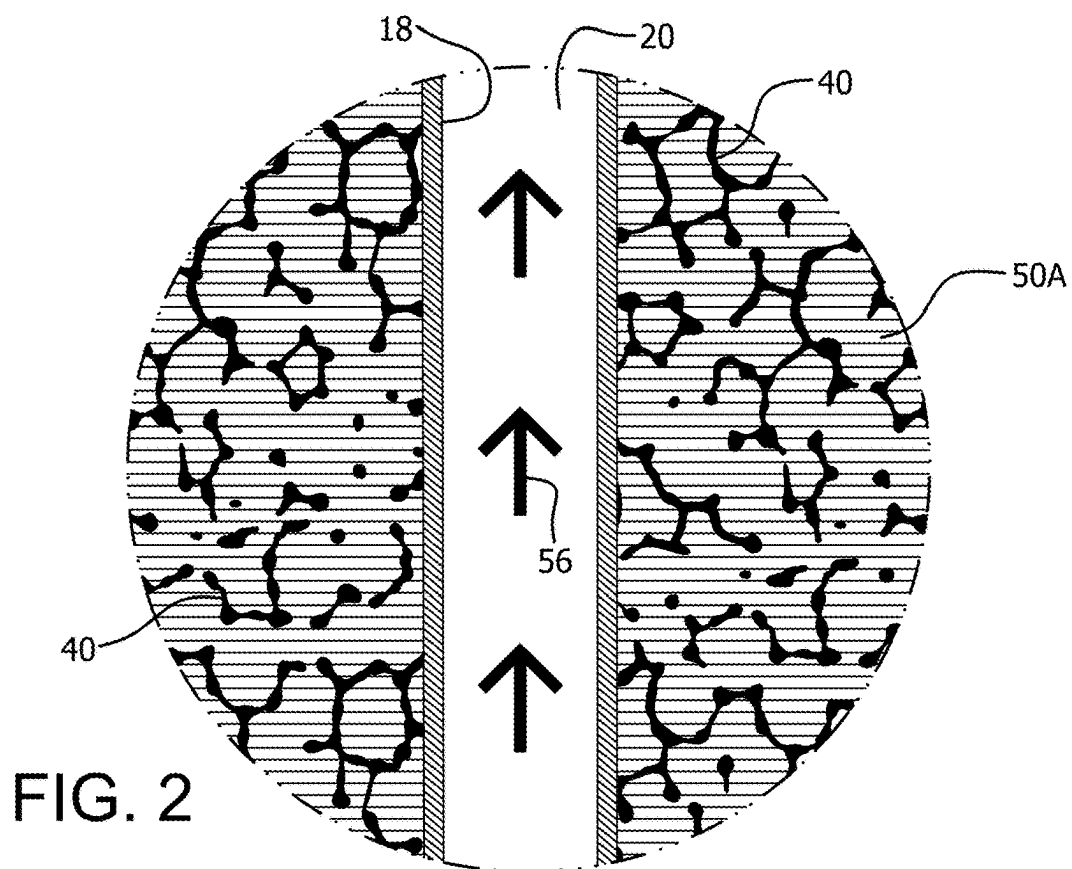


FIG. 1



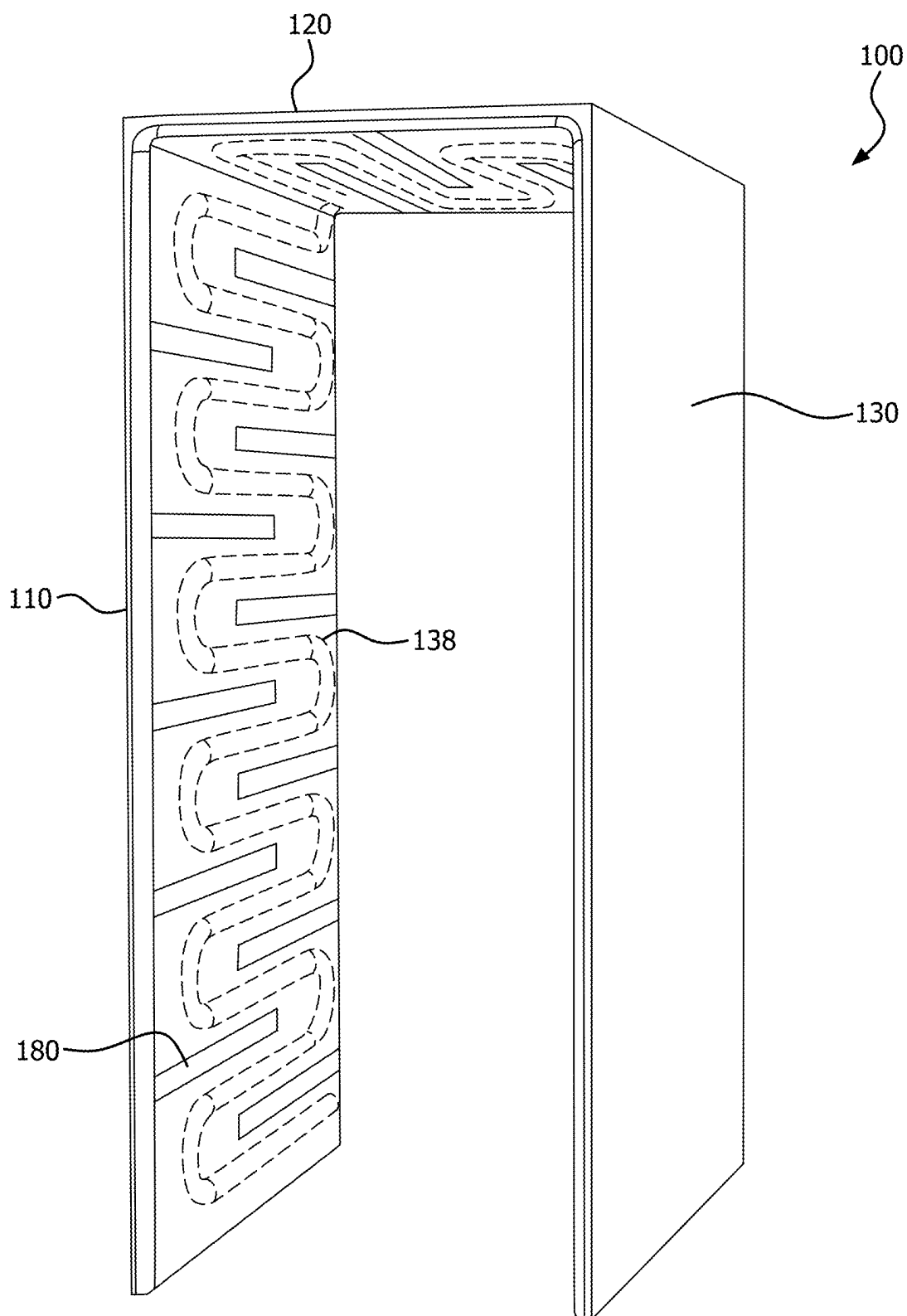
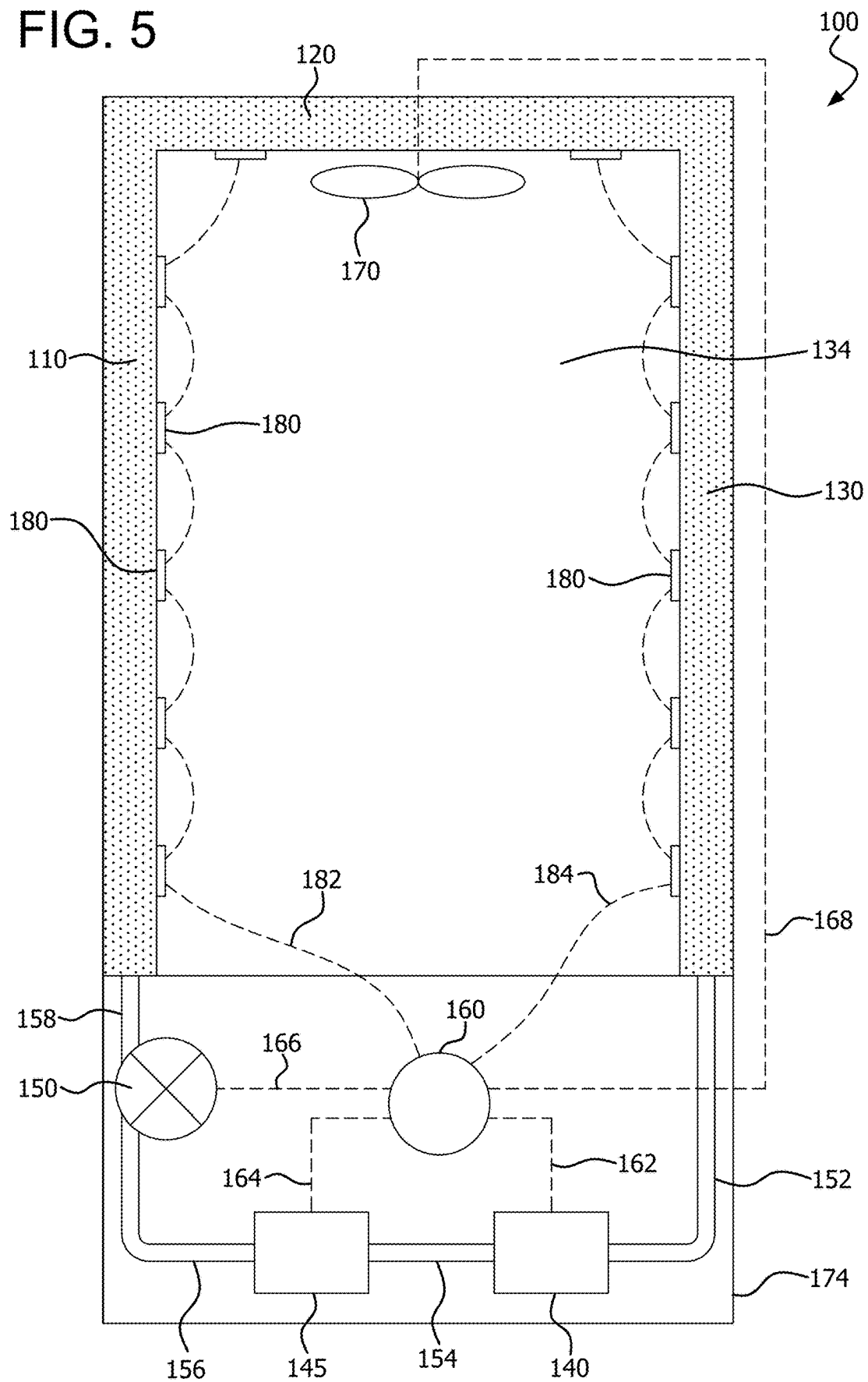


FIG. 4

FIG. 5



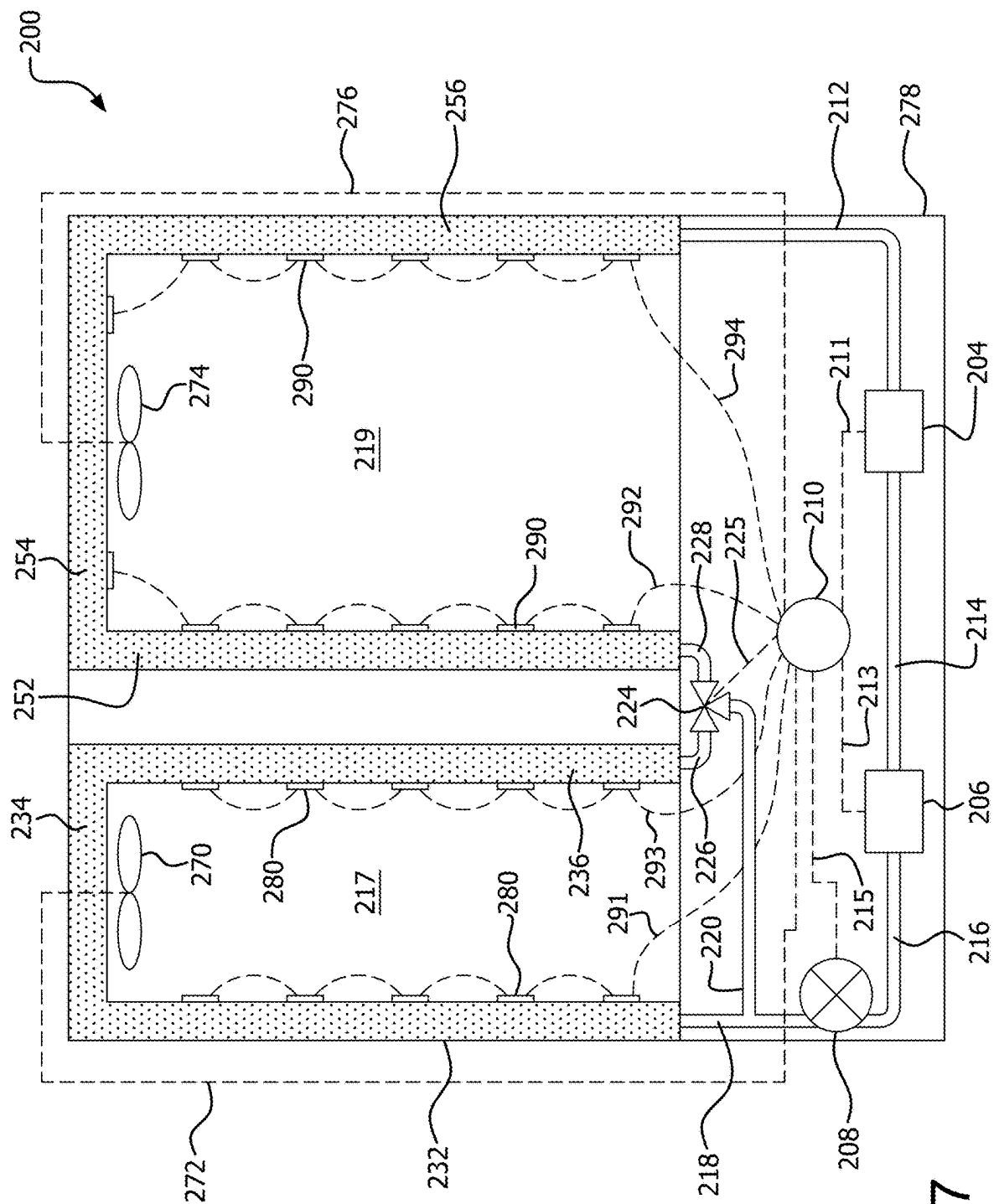


FIG. 7

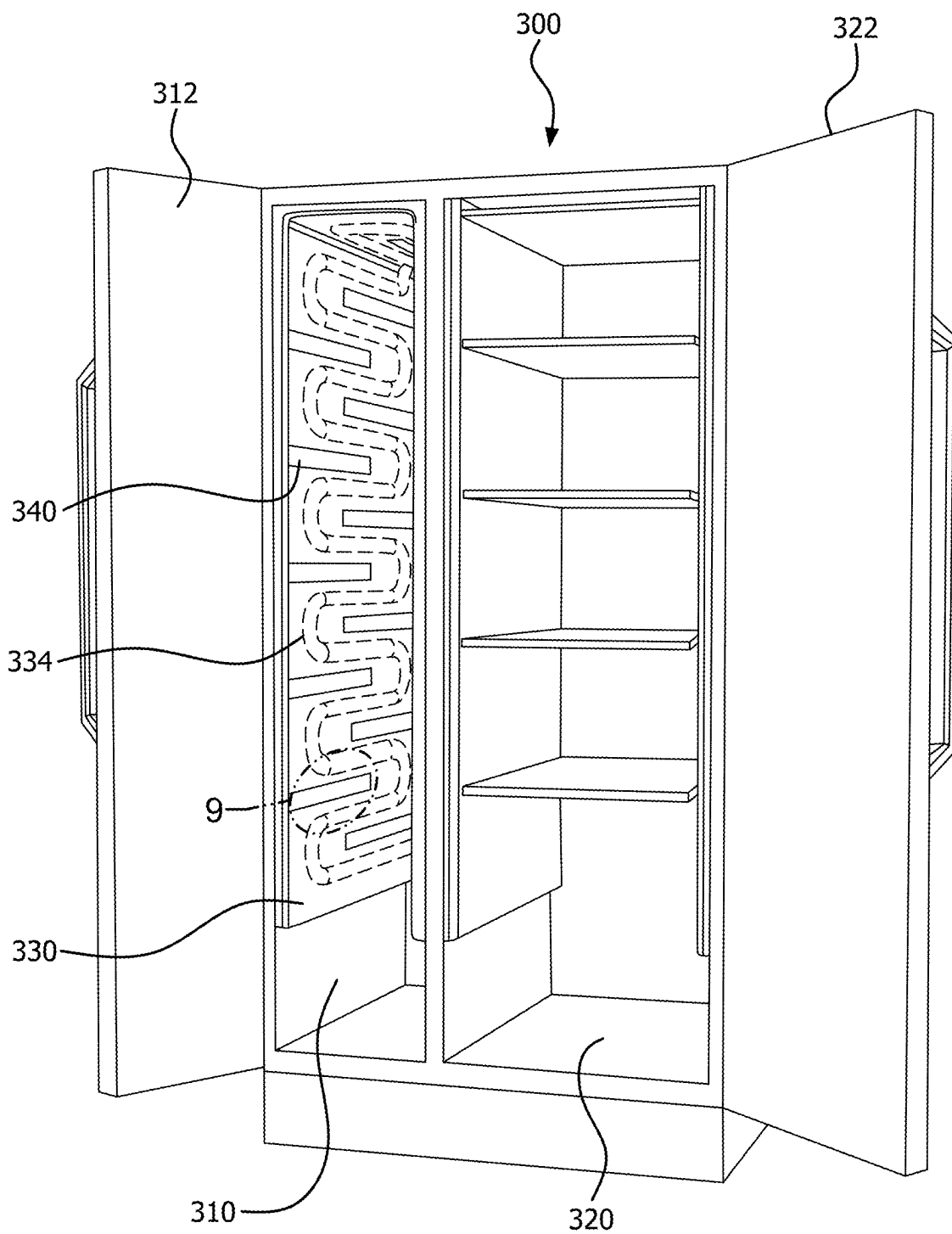


FIG. 8

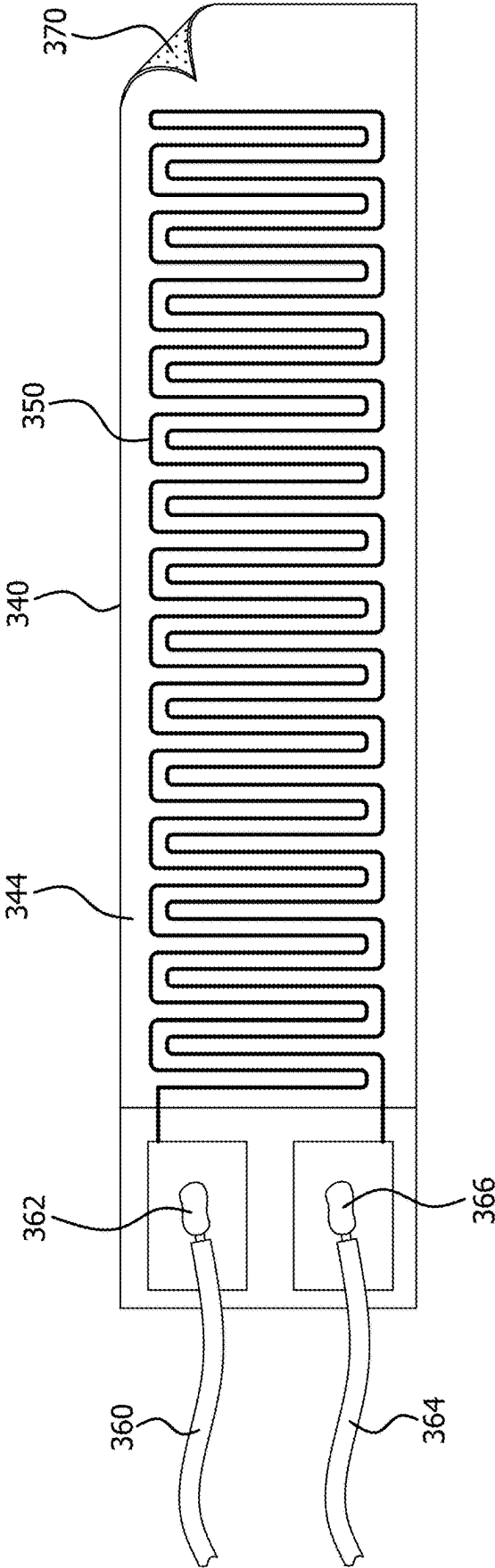


FIG. 9

REFRIGERATION SYSTEM AND METHOD ENABLING DEMAND FLEXIBILITY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to 63/557,625, filed on Feb. 26, 2024, entitled “REFRIGERATOR WITH COLD ENERGY STORAGE ENABLING DEMAND FLEXIBILITY”, the entire disclosure of which incorporated herein by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH AND DEVELOPMENT

[0002] This invention was made with government support under Contract No. DE-AC05-00OR22725 awarded by the United States Department of Energy. The government has certain rights in this invention.

FIELD OF THE INVENTION

[0003] The present invention relates to cold storage systems and methods, and more particularly to such systems and methods with demand flexibility.

BACKGROUND OF THE INVENTION

[0004] Electrical appliances account for approximately 15% of residential building primary energy consumption. Refrigerators, mostly refrigerator-freezer units, are one of the most common electrical appliances in the United States and consume nearly 7% of the home electricity. Typical household refrigerators consume 1.5-2.0 kWh of electricity per day. More than 120 million refrigerators are used across the United States, resulting in significant primary energy consumption and carbon emissions.

SUMMARY OF THE INVENTION

[0005] A refrigeration system includes a refrigeration compartment and a phase change evaporator in thermal contact with the refrigeration compartment. The phase change evaporator comprises a housing containing an open cell heat conductive metal foam and a phase change material embedded within the cells of the open cell heat conductive metal foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam. The refrigerant conduit contains or is capable of containing a refrigerant fluid. A compressor and a condenser are in refrigerant fluid communication with the refrigerant conduit of the phase change evaporator. A fan is designated for enforced heat transfer between the phase change evaporator and the refrigeration compartment air.

[0006] A target temperature can be provided for the refrigeration compartment. The phase change material can have a phase change temperature within $\pm 5^\circ$ C. of the target temperature.

[0007] The refrigeration system can be a freezer/refrigerator comprising a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change evaporator. The freezer phase change evaporator in the freezer refrigeration compartment can have a freezer phase change material in the open cell heat conductive metal foam of the freezer phase change evaporator. The fresh food compart-

ment can have a fresh food phase change material in the open cell heat conductive metal foam of the fresh food phase change evaporator.

[0008] The freezer refrigeration compartment can include a fan designated for heat transfer between the freezer phase change evaporator and the freezer refrigeration compartment, and the fresh food compartment comprises a fan designated for heat transfer between the fresh food phase change evaporator and the fresh food compartment. A freezer target temperature can be provided for the freezer refrigeration compartment. The freezer phase change material can have a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature. A fresh food target temperature can be provided for the fresh food refrigeration compartment. The fresh food phase change material has a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature.

[0009] The freezer phase change evaporator can include a freezer refrigerant conduit and the fresh food phase change evaporator can include a fresh food refrigerant conduit. Circulating refrigerant can pass first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit.

[0010] A three-way control valve can be provided. The three-way valve can be configured to switch operation between possible two modes, wherein in the first mode of operation, refrigerant flows through the refrigerant conduit of the freezer phase change evaporator and the three-way valve into the refrigerant conduit of the fresh food phase change evaporator, and in the second mode of operation, refrigerant flows from the expansion device into the three-way valve, and then into the refrigerant conduit of the fresh food phase change evaporator. This will provide an individually controllable flow of refrigerant through either the refresh food phase change evaporator or both of the freezer phase change evaporator and the refresh food phase change evaporator.

[0011] The refrigeration system can include a processor for monitoring the time of day for off-peak hours, and operating the compressor only during off-peak hours. The processor can further connect to the compressor, the condenser, the expansion valve, a fan, and defrosting heaters array in the refrigeration system.

[0012] The refrigerant conduit can be provided within the evaporator unit housing. The refrigerant conduit can be affixed to an exterior surface of the phase change evaporator housing.

[0013] A defrosting heater array can be provided in the refrigeration compartment. The defrosting heater array can comprise polyimide heater arrays. The defrosting heater array can be affixed to an exterior surface of the phase change evaporator housing.

[0014] Any suitable refrigerant can be used. The refrigerant can be at least one selected from the group consisting of HFO-1234ze and HFO-1234yf. Other refrigerants are possible.

[0015] A method for refrigerating a refrigeration compartment can include the step of providing in the refrigeration compartment a phase change evaporator comprising a housing containing an open cell heat conductive foam and a phase change material embedded within the cells of the open cell heat conductive foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam, the refrigerant conduit containing a refrigerant fluid. Refrigerant

is flowed through the refrigerant conduit sufficient to change the phase of the phase change material during an active cooling cycle. The phase change material is allowed to change phase and thereby absorb heat from the refrigeration compartment through the heat conductive open cell foam during a passive cooling cycle.

[0016] The active cooling cycle can be performed only during off-peak hours. The method can include using a processor for monitoring the time of day for off-peak hours, and operating the compressor and the active cooling cycle only during off-peak hours. The method can include monitoring the temperature in the refrigeration compartment with a temperature sensor.

[0017] The method can include the step of providing a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change evaporator. The freezer phase change evaporator in the freezer refrigeration compartment can have a freezer phase change material in the open cell heat conductive metal foam of the freezer phase change evaporator. The fresh food compartment can have a fresh food phase change material in the open cell heat conductive metal foam of the fresh food phase change evaporator.

[0018] The method can include the step of providing the freezer refrigeration compartment with a freezer phase change evaporator comprising a defrosting heater array, the defrosting heater array being affixed to an exterior surface of the phase change evaporator housing. The defrosting heater array can comprise polyimide heater arrays. The polyimide heater arrays are operated to defrost the freezer refrigeration compartment.

[0019] The method can include the step of providing the fresh food refrigeration compartment with a fresh food phase change evaporator further comprising a defrosting heater array. The defrosting heater array can be affixed to an exterior surface of the phase change evaporator housing. The defrosting heater array can comprise polyimide heater arrays. The method can include the step of operating the polyimide heater arrays to defrost the fresh food refrigeration compartment.

[0020] The method can further include the step of providing a freezer target temperature for the freezer refrigeration compartment, and providing a freezer phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature. The method can include providing a fresh food target temperature for the fresh food refrigeration compartment, and providing a fresh food phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature.

[0021] The freezer phase change evaporator can comprise a freezer refrigerant conduit and the fresh food phase change evaporator can comprise a fresh food refrigerant conduit. The method can comprise the step of circulating refrigerant first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit during the active cooling cycle.

[0022] An evaporator is provided for a refrigeration system comprising a compressor, a condenser and an expansion valve. The evaporator comprises a housing containing an open cell heat conductive foam and a phase change material embedded within the cells of the open cell heat conductive foam. A refrigerant conduit is in thermal contact with the

open cell heat conductive foam. The refrigerant conduit is capable of transferring heat with the metal foam and the phase change material.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] There are shown in the drawings embodiments that are presently preferred it being understood that the invention is not limited to the arrangements and instrumentalities shown, wherein:

[0024] FIG. 1 is a perspective view of a phase change evaporator according to the invention.

[0025] FIG. 2 is an expanded breakaway view of area 2 in FIG. 1, in a first mode of operation.

[0026] FIG. 3 is an expanded breakaway view of area 2 in FIG. 1, in a second mode of operation.

[0027] FIG. 4 is a perspective view of a three-sided phase change evaporator, partially in phantom.

[0028] FIG. 5 is a front schematic depiction of the three-sided phase change evaporator of FIG. 4 as installed in a refrigeration system.

[0029] FIG. 6 is an exploded perspective view of freezer phase change evaporator and a fresh food phase change evaporator in a freezer/refrigerator refrigeration system, partially in phantom.

[0030] FIG. 7 is a front schematic depiction of the freezer phase change evaporator, and the fresh foods phase change evaporator of FIG. 6 as installed in a refrigeration system.

[0031] FIG. 8 is a perspective view of a cold storage unit, partially in phantom.

[0032] FIG. 9 is an expanded view of area 9 in FIG. 8 showing a defrosting resistive heater.

DETAILED DESCRIPTION OF THE INVENTION

[0033] A refrigeration system includes a refrigeration compartment (or multiple refrigeration compartments), a phase change evaporator (or multiple phase change evaporators), a compressor and a condenser. The phase change evaporator is in thermal contact with the refrigeration compartment, and comprises a housing with an interior space. An open cell heat conductive foam having a phase change material embedded within the cells of the open cell heat conductive foam is provided in the interior space of the housing. A refrigerant conduit is provided and is in thermal contact with the open cell heat conductive metal foam. The refrigerant conduit contains a refrigerant fluid and communicates with a refrigeration cycle conduit which circulates the refrigerant in a refrigeration cycle between the phase change evaporator, the compressor, the condenser, and an expansion valve before returning to the phase change evaporator.

[0034] The refrigeration system can be any suitable cold storage apparatus, including but not limited to a refrigerator, a freezer, a combination refrigerator and freezer, and cold storage container. The refrigeration system can be used for residential or commercial purposes. A combination refrigerator and freezer can include a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change evaporator. The freezer phase change evaporator in the freezer refrigeration compartment has a freezer phase change material in the open cell heat conductive foam of the freezer phase change evaporator, and the fresh food com-

partment has a fresh food phase change material in the open cell heat conductive foam of the fresh food phase change evaporator. The phase change materials are used to store and release large amounts of cooling energy by undergoing a phase change between their liquid and solid states. The phase change materials are selected to possess high latent heat, be chemically inert, and have no phase segregation. Various phase change materials with freezing/melting phase change temperatures ranging from 6° C. to −30° C. can be used to fill the phase change evaporator.

[0035] The refrigeration system can further have an assigned target temperature for the refrigeration compartment. The target temperature will vary with the intended use such as freezer, fresh food storage, commercial storage, and the like. The phase change material has a phase change temperature within $\pm 5^\circ$ C. of the target temperature. A freezer target temperature can be provided for the freezer refrigeration compartment. The freezer phase change material can have a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature. A fresh food target temperature can be provided for the fresh food refrigeration compartment. The fresh food phase change material can have a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature. The freezer phase change material can be the same or a different phase change material than the fresh food phase change material.

[0036] Different requirements of temperature controls in the freezer and fresh food compartments require different and optimal phase change materials in each phase change evaporator to maintain proper temperatures in these compartments. Example of two suitable phase change materials, which are listed in Table 1, can store cold energy as latent heat at nearly isothermal temperatures of 4° C. and −20° C. for PCMs A and B, which are paraffin waxes, respectively,

[0037] The phase change materials adopted have consistent, repeatable performance over thousands of thermal melt/solidify cycles. The phase change materials can be produced from renewable agricultural sources and are readily biodegradable and nontoxic. Other examples of suitable phase change material include fatty acids and non-paraffin organics (including esters and alcohols).

[0038] The refrigerant is the fluid which is circulated through the compressor, condenser, expansion valve, and evaporator according to known refrigeration principles and stion principles and systems. Many different refrigeration cycle refrigerants are known which are suitable for the invention. The refrigerant comprises at least one selected from the group consisting of R134a, HFO-1234ze and HFO-1234yf. R134a is a conventional hydrofluorocarbon refrigerant used in household refrigerators, but it has a high global warming potential (GWP) of 1,430. Low-GWP refrigerants are preferred, such as HFO-1234ze or HFO-1234yf, or emerging low-GWP refrigerants as a drop-in replacement for R134a. HFO-1234ze and HFO-1234yf have unique advantages, such as zero ozone depletion potential (ODP) and very small GWPs, have a cooling capacity and performance comparable with R134a, and are easily converted from the R134a design system with minimal changes. HFO-1234ze and HFO-1234yf enable 8.1% and 5.5%, respectively, CO₂ emissions reduction for life cycle climate performance (LCCP), as shown in Table 2.

TABLE 2

Comparison of GWP and LCCP for R134a, HFO-1234ze, and HFO-1234yf.						
Refrigerant	GWP	GWP reduction	ASHRAE safety class	Flammability limits at 20° C.	Flammability limits at 100° C.	LCCP CO ₂ e emissions reduction
R134a	1,430	—	A1	Nonflammable	Nonflammable	Baseline
HFO-1234ze	6	>99%	AL2	Nonflammable	7.0-12.0 vol %	8.1%
HFO-1234yf	4	>99%	AL2	Nonflammable	6.2-12.3 vol %	5.5%

which match common set point temperature controls for freezer and fresh food compartments.

TABLE 1

Properties of selected PCMs in the cooling energy storage system.				
	Heat storage capacity	Peak melting point	Liquid thermal conductivity	Liquid density
PCM A for freezer compartment	240 J/g	−20° C.	0.55 W/mK	1.06 g/mL
PCM B for fresh food compartment	187 J/g	4° C.	0.15 W/mK	0.88 /mL

[0039] In addition, HFO-1234ze and HFO-1234yf have an atmospheric lifetime of only 11 days, whereas R134a and CO₂ have atmospheric lifetimes of 13 years and over 500 years, respectively. Thus, replacing R134a with HFO-1234ze or HFO-1234yf in refrigerators can enable additional greenhouse gas (GHG) emissions reduction.

[0040] The heat conductive metal foam can be any of several materials. Metal foam is a lightweight, porous material made by incorporating gas-filled pores into a metal matrix, and these pores can be open-cell (interconnected) or closed-cell (sealed). Open-cell metal foam materials are used in the invention. Open-cell metal foam materials combine the advantages of metal properties with the benefits of highly permeable lightweight materials, offering open porosity, low relative density, high cell edge thermal conductivity, and a large accessible surface area per unit vol-

ume. These features make open-cell metal foams excellent for improving phase change material heat conduction to develop efficient, compact, and lightweight phase change material energy storage components. Open-cell metal foams can substantially improve the heating/cooling storage/retrieval rate of phase change materials because the foams enhance heat conduction and have a large network surface area. Metal foams that are suitable for use with the invention can be open-cell aluminum foam with a 5.8 W/m-K bulk thermal conductivity, high porosity of from 90% to 96% porosity, and a low density of from 4% to 10% of solid aluminum. This foam suitably increases the effective thermal conductivity of the composite phase change materials through the large interface between metal foam and phase change material filling to create a composite material. Table 3 lists the key features of the aluminum foam.

TABLE 3

Physical features of aluminum foam (8% nominal density.)		
Bulk thermal conductivity	Coefficient of thermal expansion	Specific surface area
5.8 W/(m · K)	23.6×10^{-6} m/(m · K)	45 m ² /m ³

[0041] The phase change material evaporators with the metal foam reduce the freezing or melting time of the phase change material by 50%-60% compared to a phase change material without the metal foam.

[0042] The open-cell metal foam substantially improves the heating/cooling storage and retrieval rates of phase change materials in the phase change evaporators because the foams enhance heat conduction and have a large network surface area. The open-cell metal form combines the advantages of metal properties with the benefits of highly permeable lightweight materials, offering open porosity, low relative density, high cell edge thermal conductivity, and a large accessible surface area by unit volume. These features make open-cell metal foams excellent for improving phase change material heat conduction.

[0043] The open cell heat conductive foam and phase change material are provided within the open interior of the phase change evaporator housing. The refrigerant conduit can also be provided within the phase change evaporator housing. The refrigerant conduit can alternatively be affixed to an exterior surface of the phase change evaporator housing for ease of manufacturing, in thermal contact with the open cell metal foam and phase change material through the walls of the refrigerant conduit and the phase change evaporator.

[0044] The freezer phase change evaporator can comprise a freezer refrigerant conduit and the fresh food phase change evaporator can comprise a fresh food refrigerant conduit. The circulating refrigerant can pass first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit. Alternatively, the fresh food refrigerant conduit and the freezer refrigerant conduit can be arranged in parallel. The heat transfer surface area of the freezer refrigerant conduit can in some instances exceed the heat transfer

surface area of the fresh food refrigerant conduit to meet the lower freezer target temperature of the freezer phase change evaporator and thereby more significant cooling demands of the freezer compartment. In other instances, the fresh food refrigerant conduit may require a greater heat transfer surface than the freezer refrigerant conduit due to the larger volume serviced in the fresh foods compartment. The heat transfer surface area of the respective refrigerant conduit must be adjusted to meet the demands of the application. For example, the refrigerant conduit can be formed to take a tortuous or serpentine path through the metal foam to increase the thermal contact between the refrigerant contact and the metal foam, and thereby also with the phase change material.

[0045] The refrigeration system can further comprise a processor for monitoring the temperature and operating one or more of the compressor, the condenser, the phase change evaporators, or fans designed to move air across the phase change evaporators. The processor can also track the time of day for off-peak hours, and operate the compressor only during off-peak hours.

[0046] Fans can be added to improve air movement and heat transfer between air in the refrigeration unit and the phase change evaporators in the refrigeration compartments. The processor can be used to control the fans and compressor operation to realize flexible load demand management.

[0047] The refrigeration unit can further comprise a defrosting heater array in the refrigeration compartment. The defrosting heater array can include comprises polyimide heater arrays. The defrosting heater array can be affixed to the phase change evaporator housing. Defrosting is essential for household refrigerators after 8-12 h of operation to avoid significantly degrading performance. Defrosting using an electric heater is the prevailing method for removing frost in frost-free refrigerators. An electric heater is usually installed at the bottom of the evaporator in the freezer compartment that transfers heat to the frost through an upward natural convection of the heater-generated warm air. The air temperature around the heater is typically heated to over 40° C., whereas the air temperature in the freezer compartment is approximately -18° C.~-20° C.. Increasing the temperature from -18° C.~-20° C. to 40° C. requires a substantial electrical power input and significant cold energy loss. Experimental data show that the defrost efficiency, e.g., the ratio between the ideal electrical-heating energy needed to remove frost and the energy actually released by the heater, is only 30.3%. Thus, an efficient direct-contact defrosting approach is used with a polyimide heater array that can be mounted intermittently and uniformly by applying pressure-sensitive adhesive or self-fusing tape on the phase change evaporator housing surface. The polyimide heater array directly heats the heat transfer surface of the phase change material evaporator and frost accumulated on the surface instead of heating the air. Consequently, it can substantially reduce the electrical energy consumption in defrosting.

[0048] The adopted polyimide heating elements are a thin, etched-foil circuit laminated between two lightweight polyimide films that provide high dielectric strength, superior chemical resistance, and low outgas characteristics, as shown in Table 4.

TABLE 4

Polyimide heater specifications used in direct contact defrosting.					
Size	Standard thickness	Maximum watt density	Operating temperature	Dielectric strength	Cost
Up to 0.91 m.	0.18 mm	7.8 W/cm ²	-50° C.~204° C.	1,000 VAC	\$47.4/m ²

[0049] In addition to being lightweight, the polyimide flexible defrosting heaters provide other advantages, such as (1) precise heat distribution and zoning with intricate etched foil heating elements; (2) an ultrathin profile at a typical 0.18 mm value that promotes efficient heat transfer; (3) factors that make them ideal for extreme temperature environments, such as -40° C.; and (4) a semitransparent film that allows parts and internal structures to be visually inspected easily. The polyimide heaters can provide a fast and efficient defrosting heater solution, enabling precise heat distribution for defrosting.

[0050] Precise temperature control using the polyimide heaters can be obtained with the use of temperature sensors. For example, two temperature sensors—one mounted to the heated section and one mounted to the unheated section of the phase change evaporator housing surface, can monitor the temperature variations and apply an electronic temperature controller that cycles the defrosting heater at a set temperature to achieve optimal defrosting. The refrigeration system can include a temperature sensor. This controllable heat achieves optimal defrosting within a narrow temperature range, e.g., 4° C.-5° C., enabling a very efficient uniform heating response and temperature control throughout the overall defrosting surface. As a result, substantial energy savings compared with conventional electric heater defrosting can be accomplished, further boosting refrigerator efficiency.

[0051] A method for refrigerating a refrigeration compartment includes the step of providing a phase change evaporator in the refrigeration compartment. The phase change evaporator comprises a housing containing an open cell heat conductive foam and a phase change material embedded within the cells of the open cell heat conductive foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam. The refrigerant conduit contains a refrigerant fluid. Refrigerant is flowed through the refrigerant conduit sufficient to change the phase of the phase change material during an active cooling cycle. The phase change material is allowed to change phase and thereby absorb heat from the refrigeration compartment through the heat conductive open cell foam during a passive cooling cycle.

[0052] The method can include performing the active cooling cycle only during off-peak hours. The processor can be used for monitoring the time of day for off-peak hours, and operating the compressor and the active cooling cycle only during off-peak hours. The term off-peak hours can vary depending on sources of power (wind, solar, fossil fuel) and times of peak demand, which can vary over the course of the year. The processor can be used to define the hours that the refrigeration cycle can operate, and to adjust the hours of refrigeration cycle operation based upon these

factors. The method can also include the step of monitoring the temperature in the refrigeration compartment with one or more temperature sensors.

[0053] The refrigeration system can be a refrigerator/freezer, and the method can include the step of providing a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change evaporator. The freezer phase change evaporator in the freezer refrigeration compartment can have a freezer phase change material in the open cell heat conductive foam of the freezer phase change evaporator, and the fresh food compartment can have a fresh food phase change material in the open cell heat conductive foam of the fresh food phase change evaporator. The method can further include the step of providing a freezer target temperature for the freezer refrigeration compartment, and providing a freezer phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature, and providing a fresh food target temperature for the fresh food refrigeration compartment, and providing a fresh food phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature.

[0054] The freezer phase change evaporator can include a freezer refrigerant conduit and the fresh food phase change evaporator comprises a fresh food refrigerant conduit. The method can include the step of circulating refrigerant first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit during the refrigeration cycle.

[0055] The invention provides a household refrigerator that can achieve an energy consumption reduction of at least 20% based on a conventional refrigerator having frozen and fresh food compartments. Such a model refrigerator includes (1) two phase change evaporators filled with appropriate phase change materials placed at frozen and fresh food compartments, respectively, achieving long-duration cold energy storage; (2) phase change material heat conduction enhanced by using open-cell metal foam; (3) a direct-contact defrosting approach using a polyimide heater array that enables efficient defrosting at low-power consumption; and (4) a low-GWP refrigerant that enables more than a 99% reduction of direct refrigerant GWP. A suitable processor provides refrigerator operation control. The refrigerator stores cold energy in the phase change material during the night or other periods of excess power availability while maintaining the freezer and fresh food compartments at appropriate temperatures. During the day, cold energy stored in the phase change material is used to maintain the temperature of the two compartments without turning on the compressor. The single on/off cycle minimizes compressor on/off loss, extends steady-state operation at peak design efficiency, and achieves transformational 100% load shedding during the period of peak electricity demand. The direct-contact defrosting approach directly heats the frost

accumulated on the surface instead of heating the air. It enables precise defrosting and substantially reduces electrical power need. Low-GWP refrigerants are considered to replace R134a, enabling additional GHG emissions reduction. Integrating the disclosed technologies allows the new refrigerator to achieve an overall 30% reduction in CO₂ emissions.

[0056] There is shown in FIGS. 1-3 a phase change evaporator 10. The phase change evaporator 10 has a housing 14 containing a refrigerant conduit 18 with an open interior 20 and an inlet 22 and an outlet 26. A port 30 is provided to introduce and remove phase change material from the housing 14. As shown in FIG. 2, a conductive open cell metal foam 40 is provided in the housing 14 and is in thermal contact with the refrigerant conduit 18. A phase change material 50 is provided in the housing 14 and fills the interstitial spaces of the open cell metal foam 40. As shown in FIG. 2, the phase change material 50A is in a first phase such as a solid phase as refrigerant 56 flows through the refrigerant conduit 18 and removes heat from the phase change material 50A through thermal contact with the open cell metal foam 40 and the refrigerant conduit 18. As shown in FIG. 3, during times when the refrigeration cycle is off and refrigerant flow ceases, as during daylight hours or when power is at a premium, the phase change material absorbs heat shown as arrow 60 from the refrigeration compartment and changes phase from the first phase to a second phase such as a liquid phase 50B.

[0057] There is shown in FIG. 4 a three-sided phase change evaporator 100 such as would be used for a refrigerator or other cold storage compartment. The phase change evaporator 100 includes a first side 110, a second, top side 120, and a third side 130. A refrigerant conduit 138 runs through the phase change evaporator 100. Resistive defrosting heaters 180 are provided on interior surfaces of the phase change evaporator 100.

[0058] The phase change evaporator 100 is shown schematically in a refrigerator or other refrigeration device in FIG. 5. The refrigerator system includes a compressor 140, a condenser 145, and an expansion valve 150. A refrigerant conduit 152 leads from the phase change evaporator 100 to the compressor 140. A refrigerant conduit 154 leads from the compressor 140 to the condenser 145. A refrigerant conduit 156 leads from the condenser 145 to the expansion valve 150. Another refrigerant conduit 158 leads from the expansion valve 150 to the refrigerant conduit 138 of the phase change evaporator 100.

[0059] A processor 160 can be provided to direct operation of the refrigeration system. For example, the processor 160 can be connected by a communication line 162 to the compressor 140. The processor 160 can be connected through a communication line 164 to the condenser 145. The processor 160 can be connected to the expansion valve through a communication line 166. The processor 160 can be connected to a fan 170 through a communication line 168. The processor 160 can be connected to defrosting heaters 180 through a communication line 182. A second communication line 184 can be provided to afford more specific control of resistive heaters 180 in the refrigeration compartment 134. More circuits can be provided to enable more localized control of defrosting. Additional lines of communication to additional fans 170 and defrosting heaters 180 can be provided to perform more specific localized control of cooling and defrosting heating.

[0060] There is shown in FIG. 6 a refrigeration system 200. The refrigeration system 200 has an outer shell 202 and includes a first freezer compartment 217 with a door 216 and a fresh food compartment 219 with a door 220. A freezer phase change evaporator 230 has a first side 232, a second top side 234, and a third side 236. A refrigerant conduit 240 can be provided in the interior of the freezer phase change evaporator 230 and thereby is in thermal contact with open cell metal foam and freezer phase change material which fills the freezer phase change evaporator 230.

[0061] The refrigeration system 200 also includes fresh food phase change evaporator 250 for the fresh food compartment 219. The fresh foods phase change evaporator 250 includes a first side 252, a second top side 254, and the third side 256. A refrigerant conduit 260 is provided within the fresh foods phase change evaporator 250 and is in thermal contact with an open cell foam and phase change material which fills the fresh foods phase change evaporator 250. Defrosting heaters 290 can be provided on a surface of the fresh food phase change evaporator.

[0062] A schematic depiction of the refrigeration system 200 shown in FIG. 7 includes a compressor 204, a condenser 206, and an expansion valve 208. A refrigerant conduit 212 delivers refrigerant from the fresh food phase change evaporator 250 to the compressor 204. A refrigerant conduit 214 delivers refrigerant from the compressor 204 to the condenser 206. A refrigerant conduit 216 delivers refrigerant from the condenser 206 to the expansion valve 208. A refrigerant conduit 218 delivers refrigerant from the expansion valve 208 to the freezer phase change evaporator 230. A branch refrigerant conduit 220 leads from the expansion valve 208 to a three-way control valve 224. A refrigerant conduit 226 leads from the refrigerant conduit 240 of the freezer phase change evaporator 230 to the three-way valve 224. A refrigerant conduit 228 leads from the three-way valve 224 to the refrigerant conduit 260 of the fresh food phase change evaporator 250.

[0063] The three-way valve 224 is capable of switching operation between possible 1st and 2nd modes. In the first mode of operation refrigerant flows through the refrigerant conduit 240 of the freezer phase change evaporator 230, the refrigerant conduit 226, the three-way valve 224, and the refrigerant conduit 228 into the refrigerant conduit 260 of the fresh food phase change evaporator 250. This is essentially a series flow. In the second mode of operation, refrigerant flows through the branch refrigerant conduit 220 into the three-way valve 224, and through the refrigerant conduit 228 into the refrigerant conduit 260 of the fresh food phase change evaporator 250. The three-way valve 224 prevents flow from the refrigerant conduit 240 of the freezer phase change evaporator 230. It can provide a parallel and individually controllable flow of refrigerant through either the refresh food phase change evaporator 250 or both of the freezer phase change evaporator 230 and the refresh food phase change evaporator 250.

[0064] The refrigeration system 200 can also include a processor 210 to control operation of the refrigeration system 200. The processor 210 can, for example control the compressor 204 through a control line 211. A control line 213 permits control of the condenser 206 by the processor 210. A control line 215 connects the processor 210 to the expansion valve 208. A control line 225 connects the processor 210 to the valve 224. A control line 272 connects the processor 210 to a fan 270 in the freezer compartment 217.

A control line **276** connects the processor **210** to a fan **274** in the fresh foods compartment **219**. A control line **291** connects the processor **210** to a first series of defrosting heaters **280** in the freezer compartment **217**. A second control line **293** connects a second series of the resistive heaters **280** in the freezer compartment **217** to the processor **210**. A control line **292** connects the processor **210** to a first series of resistive heaters **290** in the fresh foods compartment **219**. A control line **294** connects the processor **210** to a second series of resistive heaters **290** in the freezer compartment **219**.

[0065] The evaporator system shown in FIGS. 6-7 with phase change material cooling energy storage includes two phase change evaporators: one in the freezer compartment and one in the fresh food compartment. Both phase change evaporators are designed as U-type phase change evaporator slabs, the insides of which are filled with phase change material and an open cell metal foam material that improves phase change material heat conduction. In the phase change evaporator of the freezer compartment, the surface of the phase change evaporator housing can be covered with a serpentine cold refrigerant conduit that is in contact with the air so that air can be cooled quickly to achieve a -17.8°C . (0°F .) freezer compartment temperature, particularly after defrosting. Additionally, cold refrigerant that moves through the serpentine refrigerant conduit transfers cold energy to the phase change material for cooling use during periods when the compressor is off. Considering no defrosting requirements and the temperature set at 3.9°C . (39°F .) in the fresh food compartment, the surface without a refrigerant conduit in the phase change evaporator is in contact with the air, so cold refrigerant in the serpentine refrigerant conduit transfers cold energy to the phase change material, which cools the air. In one embodiment, refrigerant flow exiting the freezer compartment is transferred to the fresh food phase change evaporator in the freezer compartment and then flows to the phase change evaporator in the fresh food compartment.

[0066] There is shown in FIG. 8 a cold storage system **300** in which a low temperature storage compartment **310** has door **312** and a higher temperature storage compartment **320** has a door **322**. The lower temperature storage compartment **310** has a lower temperature phase change evaporator **330** provided therein. The lower temperature phase change evaporator **330** can have a refrigerant conduit **334** within and in thermal contact with an open cell metal foam and lower temperature phase change material embedded in the open cells of the metal foam. The higher temperature storage compartment **320** in this embodiment does not possess a phase change evaporator due to the reduced thermal load required by this higher temperature compartment. The refrigeration system **300** demonstrates that phase change evaporators according to the invention can be provided as-needed in one but not all refrigeration compartments.

[0067] The cold storage phase change evaporator **330** can be fitted with defrosting resistive heaters **340**. There is shown in FIG. 9 such a heater **340** which has a flexible base **344** and a resistive heating conductor **350** secured to the flexible base **344**. A conductor **360** and electrical contact **362** and a second electrical conductor **364** and associated contact **366** supply power to the defrosting resistive heater **340** and can be controlled by a processor as previously described. The heater **340** can be secured by suitable means such as an adhesive **370** on a back side of the flexible base **344**.

[0068] The disclosed household refrigerator uses advanced evaporators with phase change material based long-duration cold energy storage, phase change material heat conduction enhancement using a metal foam material, direct-contact defrosting technology, and a low-global warming potential (GWP) alternative refrigerant to achieve flexible load demand management and transformational efficiency improvement in excess of 20%. This enables a 30% reduction in CO_2 emissions and a nearly 100% load shift from daytime to nighttime operation. Replacing all conventional refrigerators in homes and commercial buildings with the disclosed refrigerator could save up to 167 TBtu of primary energy consumption and reduce CO_2 emissions by 7.2 million t. This replacement would directly support a smooth transition toward meeting the future goals for net-zero carbon emissions. Additionally, the disclosed refrigerator's load shifting capability enables approximately 0.7%-1.0% of daytime electricity demand to be shifted to nighttime operation nationwide.

[0069] The phase change evaporator housing can be made of various materials and sizes depending on the application. The phase change evaporator housing can be made of a metal, including but not limited to aluminum, copper, and steel.

[0070] A fabricated sample was prepared which measured 1.14 cm long, 40.0 cm wide, and 3.0 cm deep. The samples were installed into a refrigerator. The designed refrigeration system consisted of a compressor, a condenser, one or two phase change material evaporators, and a thermal expansion valve. The configurations of the refrigeration cycle and compartments for the freezer and refrigerator included frozen-and fresh-food compartments, respectively. In the designed refrigeration systems, the phase change material evaporators were installed along the compartment walls.

[0071] In the freezer, the compressor boosts the pressure and temperature of the refrigerant. In the condenser, the refrigerant releases heat and condenses into a liquid. The expansion valve then reduces the pressure and temperature of the refrigerant; and the phase change material evaporator, where the refrigerant absorbs heat and evaporates, transfers cooling energy to the phase change evaporators and thereby the phase change material and metal foam, thereby storing the cooling energy in the phase change material. The metal foam enhances heat conduction between the refrigerant conduit and the phase change material. The stored cooling energy can be used by managing the fan on/off command to discharge the cooling energy to the compartment air. The charging demand of cooling energy is based on phase change material status, and the charging demand of cooling energy is based on the compartment air temperature setting.

[0072] In a refrigerator/freezer with frozen and fresh-food compartments, a three-way valve can be added for flexibly managing the refrigerant flow to achieve appropriate cooling charging to the phase change materials with the phase change evaporators. The three-way valve allows the refrigerant either to flow through the evaporator coils of the phase change evaporator in sequence or to bypass the refrigerant conduit of the freezer phase change evaporator and directly enter the refrigerant conduit of the fresh food phase change evaporator in the fresh-food compartment. The cooling charging in the phase change evaporators and the discharging to compartment air are the same process in both the frozen and fresh-food compartments.

[0073] The refrigeration system can be used in fields such as manufacturing, energy, and utilities. The invention can be used for producing refrigerators, phase change materials, long-duration cold energy storage, advanced phase change evaporators with phase change material cold energy storage, or phase change material heat conduction enhancement using open-cell metal foam. Also, for using direct-contact defrosting using polyimide heater elements.

[0074] The invention can achieve flexible load demand management and transformational efficiency improvement in excess of 20% compared with a conventional baseline household refrigerator while satisfying the required cooling capacity of the refrigerator. The disclosed refrigerator can achieve a 30% reduction in direct and indirect CO₂ emissions and a nearly 100% load shift from daytime to nighttime operations. These objectives can be accomplished by a single on/off cycle that enables shifting 100% of the refrigerator daytime cooling load to nighttime operation instead of the frequent compressor on/off switching that occurs in conventional refrigerators. Cold energy can be stored in the compact phase change evaporator during the night, e.g., 9:00 p.m. to 9:00 a.m., or other periods of excess power availability while maintaining the freezer and fresh food compartments at appropriate different temperatures. During the day, the stored cold energy can be used to continuously maintain the temperature of the two compartments without running the compressor. The single startup of the compressor minimizes transient operation, extending the steady-state operation of the vapor compression system at peak design efficiency while achieving less wear and extending the lifetime of the refrigerator. Additionally, a direct-contact defrosting approach allows the electric heater energy consumption to be reduced by precisely applying heat to the frost accumulated on the heat transfer surface. Furthermore, a low-GWP refrigerant can be used in the disclosed refrigerator to further reduce carbon emissions.

[0075] The invention as shown in the drawings and described in detail herein disclose arrangements of elements of particular construction and configuration for illustrating preferred embodiments of structure and method of operation of the present invention. It is to be understood however, that elements of different construction and configuration and other arrangements thereof, other than those illustrated and described may be employed in accordance with the spirit of the invention, and such changes, alternations and modifications as would occur to those skilled in the art are considered to be within the scope of this invention as broadly defined in the appended claims. In addition, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting.

We claim:

1. A refrigeration system, comprising:

a refrigeration compartment;

a phase change evaporator in thermal contact with the refrigeration compartment, the phase change evaporator comprising a housing containing an open cell heat conductive metal foam and a phase change material embedded within the cells of the open cell heat conductive metal foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam, the refrigerant conduit containing a refrigerant fluid;

a compressor and a condenser in refrigerant fluid communication with the refrigerant conduit of the phase change evaporator; and,

a fan designated for enforced heat transfer between the phase change evaporator and the refrigeration compartment air.

2. The refrigeration system of claim 1, further comprising a target temperature for the refrigeration compartment, and wherein the phase change material has a phase change temperature within $\pm 5^\circ$ C. of the target temperature.

3. The refrigeration system of claim 1, wherein the refrigeration system is a freezer/refrigerator comprising a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change evaporator, the freezer phase change evaporator in the freezer refrigeration compartment having a freezer phase change material in the open cell heat conductive metal foam of the freezer phase change evaporator, and the fresh food compartment having a fresh food phase change material in the open cell heat conductive metal foam of the fresh food phase change evaporator.

4. The refrigeration system of claim 3, wherein the freezer refrigeration compartment comprises a fan designated for heat transfer between the freezer phase change evaporator and the freezer refrigeration compartment, and the fresh food compartment comprises a fan designated for heat transfer between the fresh food phase change evaporator and the fresh food compartment.

5. The refrigeration system of claim 3, comprising a freezer target temperature for the freezer refrigeration compartment, and wherein the freezer phase change material has a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature, and comprising a fresh food target temperature for the fresh food refrigeration compartment, wherein the fresh food phase change material has a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature.

6. The refrigeration system of claim 3, wherein the freezer phase change evaporator comprises a freezer refrigerant conduit and the fresh food phase change evaporator comprises a fresh food refrigerant conduit, and wherein circulating refrigerant passes first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit.

7. The refrigeration system of claim 3, comprising a three-way control valve, the three-way valve configured to switch operation between possible two modes, wherein in the first mode of operation, refrigerant flows through the refrigerant conduit of the freezer phase change evaporator and the three-way valve into the refrigerant conduit of the fresh food phase change evaporator, and in the second mode of operation, refrigerant flows from the expansion device into the three-way valve, and then into the refrigerant conduit of the fresh food phase change evaporator, providing an individually controllable flow of refrigerant through either the refresh food phase change evaporator or both of the freezer phase change evaporator and the refresh food phase change evaporator.

8. The refrigeration system of claim 1, further comprising a processor for monitoring the time of day for off-peak hours, and operating the compressor only during off-peak hours.

9. The refrigeration system of claim 8, wherein the processor further connects to the compressor, the condenser, the expansion valve, a fan, and defrosting heaters array in the refrigeration system.

10. The refrigeration system of claim 1, wherein the refrigerant conduit is provided within the evaporator unit housing.

11. The refrigeration system of claim 1, wherein the refrigerant conduit is affixed to an exterior surface of the phase change evaporator housing.

12. The refrigeration system of claim 1, further comprising a defrosting heater array in the refrigeration compartment.

13. The refrigeration system of claim 12, wherein the defrosting heater array comprises polyimide heater arrays.

14. The refrigeration system of claim 12, wherein the defrosting heater array is affixed to an exterior surface of the phase change evaporator housing.

15. The refrigeration system of claim 1, wherein the refrigerant comprises at least one selected from the group consisting of HFO-1234ze and HFO-1234yf.

16. A method for refrigerating a refrigeration compartment, comprising the steps of:

providing in the refrigeration compartment a phase change evaporator comprising a housing containing an open cell heat conductive foam and a phase change material embedded within the cells of the open cell heat conductive foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam, the refrigerant conduit containing a refrigerant fluid;

flowing refrigerant through the refrigerant conduit sufficient to change the phase of the phase change material during an active cooling cycle; and,

allowing the phase change material to change phase and thereby absorb heat from the refrigeration compartment through the heat conductive open cell foam during a passive cooling cycle.

17. The method of claim 16, wherein the active cooling cycle is performed only during off-peak hours.

18. The method of claim 17, further comprising the step of using a processor for monitoring the time of day for off-peak hours, and operating the compressor and the active cooling cycle only during off-peak hours.

19. The method of claim 16, further comprising the step of monitoring the temperature in the refrigeration compartment with a temperature sensor.

20. The method of claim 16, the method further comprising the step of providing a freezer refrigeration compartment with a freezer phase change evaporator and a fresh food refrigeration compartment with a fresh food phase change

evaporator, the freezer phase change evaporator in the freezer refrigeration compartment having a freezer phase change material in the open cell heat conductive metal foam of the freezer phase change evaporator, and the fresh food compartment having a fresh food phase change material in the open cell heat conductive metal foam of the fresh food phase change evaporator.

21. The method of claim 20, further comprising the step of providing the freezer refrigeration compartment with a freezer phase change evaporator comprising a defrosting heater array, the defrosting heater array being affixed to an exterior surface of the phase change evaporator housing, the defrosting heater array comprising polyimide heater arrays, and operating the polyimide heater arrays to defrost the freezer refrigeration compartment.

22. The method of claim 20, further comprising the step of providing the fresh food refrigeration compartment with a fresh food phase change evaporator further comprising a defrosting heater array, the defrosting heater array being affixed to an exterior surface of the phase change evaporator housing, the defrosting heater array comprising polyimide heater arrays, and operating the polyimide heater arrays to defrost the fresh food refrigeration compartment.

23. The method of claim 20, further comprising the step of providing a freezer target temperature for the freezer refrigeration compartment, and providing a freezer phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the freezer target temperature, and providing a fresh food target temperature for the fresh food refrigeration compartment, and providing a fresh food phase change material that has a phase change temperature within $\pm 5^\circ$ C. of the fresh food target temperature.

24. The method of claim 20, wherein the freezer phase change evaporator comprises a freezer refrigerant conduit and the fresh food phase change evaporator comprises a fresh food refrigerant conduit, and comprising the step of circulating refrigerant first through the freezer refrigerant conduit and then through the fresh food refrigerant conduit during the active cooling cycle.

25. An evaporator for a refrigeration system comprising a compressor, a condenser and an expansion valve, the evaporator comprising a housing containing an open cell heat conductive foam and a phase change material embedded within the cells of the open cell heat conductive foam, and a refrigerant conduit in thermal contact with the open cell heat conductive foam, the refrigerant conduit capable of transferring heat with the metal foam and the phase change material.

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